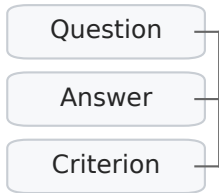


MODEL INPUT



Shared backbone
Qwen3-Reranker-0.6B

PRIMARY · TRAINING + SCORING

Hard-label head

5-class logits $z^{(h)}$

$$\mathcal{L}_h = \text{CE}(z^{(h)}, y)$$

FINAL SCORE

$\in \{1, \dots, 5\}$

AUXILIARY · TRAINING ONLY

Soft-label head

5-class logits $z^{(s)}$

$$\mathcal{L}_s = \text{CE}(z^{(s)}, d)$$

discarded at inference

JOINT TRAINING OBJECTIVE

$$\mathcal{L} = \mathcal{L}_h + \lambda \mathcal{L}_s \quad (\lambda = 1)$$